Jesus Estrada

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SUMMARY

First-year Aerospace Engineering student with a record of initiative, follow-through, and learning quickly in academic, professional, and project settings. Eager to contribute a dependable work ethic, thoughtful communication, and an adaptable mindset while continuing to grow in a collaborative organization.

EDUCATION

B.S. Aerospace Engineering **2030**

SAN JOSE STATE UNIVERSITY | San Jose, CA

- EOP Scholar.

SKILLS

Programming and Data: MATLAB, OpenRocket, Data Collection, Preliminary Data Analysis, First-Principles Analysis, Trade Studies, Technical Reporting
Test and Integration: Hands-on Testing, Functional Troubleshooting, Hardware/Software Integration, Component Verification, Design Reviews, Technical Documentation
Hardware and Systems: Computer Hardware Assembly, RFID Access-Control Hardware, Powertrain and Gear Systems, CAD Assemblies and Drawings, Component Fit and Packaging
Design and Fabrication: SolidWorks, Autodesk Inventor, Revit, Blender, 3D Printing, Cura, PrusaSlicer, Rapid Prototyping, Measurement-Based Design
Languages: English, Spanish

EXPERIENCE

Powertrain Team Member **AUG 2026 - PRESENT**

San Jose State University | Formula SAE | San Jose, CA

- Support hardware-focused powertrain development by reviewing drivetrain layout, gear ratios, component fit, packaging, and manufacturing constraints.
- Collaborate during design reviews and hardware revisions, helping identify fit, manufacturability, and integration issues before fabrication.
- Optimized gear ratio selection and analyzed powertrain performance using MATLAB simulations to enhance vehicle acceleration and efficiency.

Systems and IT Apprentice **MAY 2025 - JUN 2026**

THE SEED SCHOOL OF LOS ANGELES | Los Angeles, CA

- Diagnosed 120+ computer, projector, peripheral, and software issues using structured troubleshooting, root-cause analysis, verification, and technical documentation.
- Installed and debugged RFID access-control hardware supporting more than 150 staff members, applying hardware integration and functional-checkout procedures.
- Mentored students in hardware diagnostics, preventive maintenance, and technical problem-solving.
- Developed and implemented a standardized software deployment process for new workstations, reducing setup time by 15%. Coordinated a bi-weekly technology upgrade initiative over 12 months, deploying SCCM and PowerShell scripts to update critical software for 80+ endpoints, resulting in a 35% reduction in end-user incident tickets.

Rocket Design and Manufacturing **JAN 2025 - DEC 2025**

NCRF / CAL STATE LONG BEACH | Los Angeles, CA

- Designed and manufactured a student rocket from concept through fabrication, assembly, launch, and hands-on testing using CAD, 3D printing, and rapid prototyping.
- Used OpenRocket to model flight profiles and evaluate static stability before launch, then compared design expectations with observed performance.
- Integrated Formula One aerodynamic principles into rocket airframe design over a 9-month project, utilizing CFD tools and wind tunnel testing to achieve a 15% reduction in drag and improve launch stability by 10% compared to baseline models.

Engineering Intern Team Leader **JUN 2025 - AUG 2025**

LOS ANGELES METRO | Los Angeles, CA

- Led a 25-person team through field mapping, measurement-based design, CAD concepts, technical documentation, and evaluation of a transit redesign.
- Collected and compared project information across safety, space, budget, and schedule constraints; completed the redesign 20% ahead of schedule and presented findings to executive leadership.
- Facilitated cross-functional collaboration by integrating field data with CAD models and technical documentation for transit redesign evaluations. Spearheaded a 4-month pilot project introducing cloud-based collaboration tools to drive innovation in transit system workflows, increasing design iteration frequency by 35% and optimizing real-time data integration across a 25-person engineering team.

PROJECT

PC Builds and Support

- Built and configured 65+ PCs for 50+ paying customers using component selection, hardware assembly, operating-system installation, BIOS and driver updates, functional testing, troubleshooting, and verification.

COURSEWORK

Computer Programming for Aerospace Engineers | Aerospace Engineering | MATLAB

- Used MATLAB to write and debug scripts for engineering calculations and data analysis. Applied variables, arrays, conditional statements, loops, functions, and plotting tools to organize data, evaluate results, and solve introductory aerospace engineering problems.

Computer-Aided Design for Aerospace Engineers | Aerospace Engineering | Computer Aided Design

- Used Auto desk Inventor and Solid Works to create parametric 3D models, mechanical components, assemblies, and engineering drawings. Applied dimensions, constraints, extrusions, cuts, fillets, patterned features, section views, and technical documentation to communicate designs for manufacturing and review.

AWARDS & HONORS

AeroGreen - First Place, NCBAA Track **2026**

NCBAA SPARTAHILL HACKS

- Awarded for AeroGreen, an AI-assisted aviation-operations platform designed to help fixed-base operators coordinate aircraft arrivals, parking capacity, and service requests during periods of high demand. I helped research the operational challenge, train the AI with relevant aviation information, develop the working web prototype, and present the solution. I was the only freshman on the winning team.

International Mars Innovation Challenge **MAR 2024**

- Designed a compact water-sanitation system for space habitability, developed CAD concepts, balanced system constraints, and presented the design; placed third globally among 7,500+ entries.